

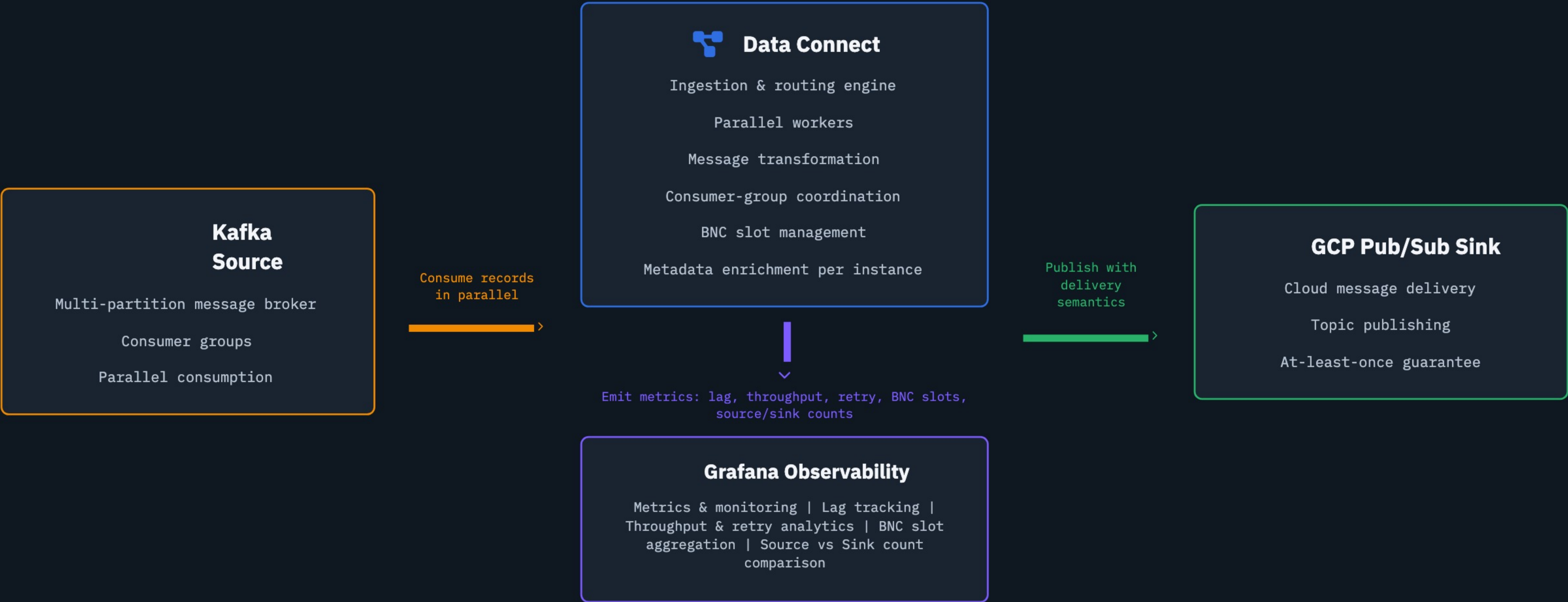
Data Connect

Data Connect Architecture

Distributed Data Pipeline – Multi Source Events
Kafka to GCP Pub/Sub

● Source ● Core ● Sink ● Observability

Container View — Data Flow Architecture



Data Flow & Observability

- **Source → Core**

[Kafka → Data Connect](#)

Data Connect uses consumer-group style workers to consume from multiple Kafka partitions in parallel, enabling horizontal scalability and high throughput ingestion.

- **Core → Sink**

[Data Connect → GCP Pub/Sub](#)

Processed messages are routed and published to Pub/Sub topics with configurable delivery semantics and reliability guarantees.

- **Core → Observability**

[Data Connect → Grafana](#)

Continuous emission of operational metrics including consumer lag, throughput rates, retry counts, BNC time-based slot data, and source/sink count aggregation comparisons for real-time monitoring, balancing verification, and alerting.

BNC — Balancing and Control

- **Source BNC Configuration**

Source can be configured to support BNC — enabling balanced consumption tracking from origin

- **Time-Based Slot Management**

Data Connect maintains time-based slots to segment and track message flow across defined intervals

- **Metadata Enrichment**

Each Data Connect instance enriches message metadata with slot information before forwarding to sink

- **Slot Data to Grafana**

Slot data is published to Grafana for real-time monitoring of slot distribution and balance health

- **Count Aggregation & Comparison**

Message counts are aggregated and compared between source and sink to verify delivery completeness and track discrepancies

Resilience, High Availability & Fault Tolerance

Resilience

- Retry mechanisms with exponential backoff
- Circuit breakers for downstream protection
- Graceful degradation under load
- Dead letter queue for failed messages
- Configurable retry policies per topic

High Availability

- Multi-instance deployment across zones
- Load balancing across workers
- No single point of failure
- Rolling deployments with zero downtime
- Health checks and readiness probes

Fault Tolerance

- Automatic failover on instance failure
- Consumer group rebalancing
- Offset management & checkpointing
- Stateless processing for easy recovery
- Redundant sink connections